DBMS Unit 1

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Database foundations

A database is an organized collection of related data. A database management system (DBMS) is software used to define, create, query, and control access to that data. A database stores the records; the DBMS manages them. A database approach can reduce unnecessary duplication and provide shared, controlled access compared with isolated files.

Worked example

A college database stores students and courses. The DBMS lets authorized staff query enrolments without maintaining separate copies in every department.

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Keys & constraints

A candidate key is a minimal set of attributes that uniquely identifies a row. A primary key is the candidate key selected to identify rows. A primary key must be unique and cannot contain NULL. A foreign key references a candidate key, commonly the primary key, in a related table. Foreign keys help enforce referential integrity. A composite key contains more than one attribute.

Worked example

Student(student_id, name) uses student_id as its primary key. Enrolment(student_id, course_id) can use both columns as a composite primary key. Enrolment.student_id references Student.student_id.

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The relational model

The relational model represents data as relations, commonly displayed as tables. A tuple is a row and an attribute is a column. The degree of a relation is its number of attributes. Its cardinality is its number of tuples. A schema describes the structure of a database; an instance is its data at a particular moment.

Worked example

A Student table with 20 rows and 3 columns has cardinality 20 and degree 3. Adding a student changes the instance, not the schema.

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Normalization basics

Normalization organizes relations to reduce avoidable redundancy and update, insertion, and deletion anomalies. In first normal form (1NF), each cell contains one atomic value for its domain, with no repeating groups. In second normal form (2NF), a relation is in 1NF and each non-prime attribute depends on the whole of every candidate key, not just part of a composite key.

Worked example

An enrolment table keyed by (student_id, course_id) should not store student_name, which depends only on student_id. Put student names in a Student relation and course enrolments in an Enrolment relation.